

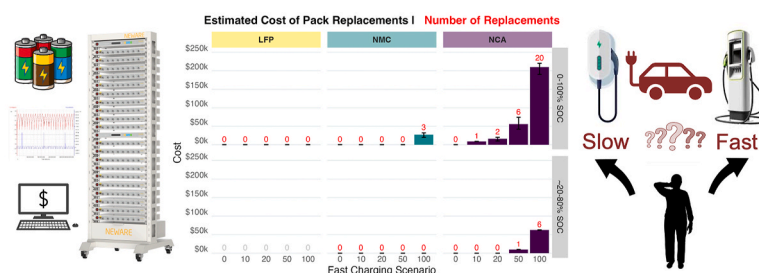
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GRAPHICAL ABSTRACT

- At equal mileage, more fast charging lowers capacity; costs vary by chemistry.
- LFP can offer lower costs for frequent fast charging with full capacity utilization.
- Cost estimates based on experimental cycling of NMC, NCA and LFP cylindrical cells.
- Different ratios of fast to slow charging events motivated by real-world behavior.
- Highlights diverse definitions of fast charging, state of health, and end of life.

Higher fast charging ratios increase costs, but it varies sharply by chemistry, with LFP showing lower costs for full capacity utilization



ABSTRACT

Keywords:
Electric vehicle
Fast charge
Fast Charging Definition
Degradation
Battery replacement
Battery lifetime
Battery costs

We investigate the relative impact of fast charging on degradation in commercial lithium-ion battery cells with electric vehicle-relevant chemistries (NMC, NCA and LFP) by cycling them under five different patterns that mix fast and slow charging events for up to 16 months with passive thermal management. We then estimate pack replacement cost to reach 150,000 lifetime vehicle miles traveled. We find that more fast-charging results in more degradation, but the magnitude of this effect varies considerably across the tested chemistries. For the most extreme scenario ($> 90\%$ fast charging, full voltage range), we find average pack replacement costs span from \$0 for LFP (no replacements) to \$27,000 for NMC (3 replacements) and \$210,000 for NCA (20 replacements). Restricting fast charging voltage to cycle $\sim 20\%$ – 80% of nameplate capacity reduces lifetime pack replacement

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costs to \$0 for NMC and \$63,000 for NCA (6 replacements). These degradation patterns suggest that when operating to limit capacity fade from frequent fast charging there are cases where the practical energy density of LFP packs can exceed that of NMC and NCA packs.

1. Introduction

Private vehicles consumed approximately 25 % of global oil (3.53 Gt of CO₂) in 2022, further encouraging decarbonization initiatives to prioritize increased and widespread battery electric vehicle (BEV) adoption [1,2]. To achieve widespread adoption, BEVs must be able to sustain frequent DC fast charger use without premature battery pack replacements, as this capability brings them closer to parity with gasoline vehicles. Frequent fast charging capabilities can accelerate BEV adoption; recent studies suggest these capabilities not only alleviate range anxiety for both urban and long-distance travel but also are increasingly preferred and integrated into daily charging routines [3–6] (SI Note 1-1). In fact, for many demographics, frequent fast charging capabilities are a prerequisite for BEV ownership, particularly for ride-share drivers and households who lack sufficient infrastructure for home charging. Regular fast charging is expected to be the primary charging method for demographics including low-income households, apartment dwellers, and those in denser urban environments. Thus, frequent fast charging capabilities are vital for widespread, increased and equitable BEV adoption.

However, there is significant concern that frequent fast charging will result in early BEV pack replacements, a substantial financial burden for any BEV owner. Fast charging has been shown to shorten battery lifetime in all major BEV battery chemistry families, including NCA (LiNi_xCo_yAl_{1-x-y}O₂), NMC (LiNi_xMn_yCo_{1-x-y}O₂) and LFP (LiFePO₄) [7–9] (SI note 1–2). While commercial LFP cells have been shown to exhibit superior cycling lifetimes compared to NCA and NMC chemistries when fast charging (higher C-rates), they remain susceptible to accelerated degradation at higher charging rates [10]. Historically, LFP has been disfavored for BEV applications due to its lower energy density.

Fast charging degradation is a much-studied topic [7–9]. There is ample documentation in the literature on how cells fail under high current applications, which can vary substantially across battery cell chemistry and are highly uncertain outside of documented experimentation (see SI note 1–2). Variation in degradation has been documented in the literature with respect to factors including, but not limited to, temperature [11], electrolyte additives [12], charging profile types (e.g. CC vs CCCV) [13], and depth of discharge cycled [14,15].

However, fast charger use is evolving. A 2024 survey by Flo, a North American BEV charging station company, found that 31 % of respondents relied on a mix of slow and fast charger stations, and 29 % did not have a dedicated charger installed in their home [16]. These behaviors are supported by increasing availability of DC fast charger stations in the U.S., up from approximately 5000 in 2019 to 34,000 in 2024 [17]. Nonetheless, there is a lack of studies that investigate degradation for fast charging mixed with slow charging across subsequent cycles. Also lacking are studies that explicitly estimate battery pack replacement costs over a vehicle’s lifetime due to different degradation regimes. Based on a thorough literature search, Table 1 summarizes (1) the few studies found to incorporate both fast and slow charger use or (2) the most relevant studies to this work that incorporate battery pack replacement cost sensitivities.

Neubauer & Wood (2015) is the only study found to have investigated the effect of a varying number of fast charging events per year. They do not report the effect on degradation rate, and the most frequent fast charging was between 90 and 100 days per year (~25–27 %). A Recurrent Auto analysis using field data found comparable degradation for BEVs fast charged more than 70 % of the time relative to BEVs only slow charged. Degradation for the frequently fast charged vehicles was lower than the authors expected. A gap for experimental studies

remains.

Modeling by Xiao et al. uses 6 months of field data from electric buses to extract an aging rate, which they use to determine battery replacement costs under slow charging, 1C, 2C and 3C charging. They estimate that fast charging (2C) led to 1–3 pack replacements, increasing reported replacement costs by 23 % [19]. Simulations by Neubauer et al. (2012) identify a cost ratio of BEVs vs conventional vehicles using a cost-optimal battery replacement schedule for 5 kW charging of packs from 16.6 to 40 kWh, finding that it is generally most cost effective not to replace the original battery before 15 years, [20]. However, in the context of car-share programs, the economics favor systems with longer range and charge strategies that foster longer battery life, lowering the likelihood that BEV ownership would be financially advantageous vs a conventional vehicle. They find economic comparisons were sensitive to drive patterns, vehicle range, and charge strategy. All charging is slow charging, but strategies vary across the maximum state of charge and between managed vs unmanaged charging times. In a subsequent study, Neubauer & Wood found considerable variation in capacity fade (~75 %–85 % retention over 10-years) and resistance growth (~15–50 % over 10-years) across different drive patterns [18]. A gap remains for estimating how pack replacement costs vary with fast charging frequency.

Describing the impact of fast charging on BEV pack replacements is

Table 1
Highlighted literature on battery replacement costs and fast/slow charging protocols (additional papers reviewed for this work can be found in SI note 1–2.).

Source	Type	Battery replacement costs	Charging protocol
Neubauer et al., 2015, SAE Technical Paper [18]	Simulation	No costs described	Fast charging events mixed with slow charging events, max 90 fast events/year
Xiao et al., 2024 Transportation Research Part D [19]	Computational model informed by field data (6 months)	Yes	Fast charging or slow charging; no mixing
Neubauer et al., 2012, J. Power Sources [20]	Simulation	No but cost ratio to conventional vehicle using a cost-optimal battery replacement schedule included	Slow charging only but multiple driving patterns
Recurrent Auto [21]	Field data analysis (~5–6 years)	No	Frequent fast charger use (>70%) compared to infrequent fast charger use (<30%)
This paper	Controlled laboratory experiment	Yes	Fast charging events mixed with slow charging events across a range of ratios

further muddled by the lack of universal definitions for “fast charging” and battery “end of life”. This, in addition to the path dependency found in battery systems, complicate predictions of battery lifetime using existing literature and underscore this paper’s concurrent experimental and modeling contribution. Table 2 summarizes notable definitions from the academic and grey literature.

Fast charging definitions vary. Units include c-rate, time for percent capacity added, charger power, time to 80 % of maximum range, and miles per charging time. In the battery literature fast charging is often defined as a full charge in 30 min or up to 2 h ($2C - C/2$); cycling faster than 10 min ($>6C$) is called extreme fast charging (XFC) [22]. In contrast, the 1996 U.S. Advanced Battery Consortium manual, defines fast charging as refilling 40 % of the battery’s nominal capacity within 15 min [23]. The National Electric Vehicle Infrastructure (NEVI) program in the United States defines fast charging by BEV DC charger power: fast charging is a charger that can provide power equal to or greater than 150 kW [24]. Older policy guidance from the California Air Resources Board (CARB) defines fast charging as a charger that provides power greater than or equal to 23 kW [25]. EVSE charger operators, car manufacturers and government agencies commonly communicate information about fast charging through time. For example, as time to 80 % of the maximum vehicle range or miles added in a given charging time [26,27]. None of these definitions reach speed parity for refilling a gas tank, which is approximately 2–5 min. Parity is approximately 936–2350 kW for a 78 kWh, 300-mile range pack like those investigated in this study (See SI Note 1–3 for calculation).

Definitions for battery end of life vary by capacity retention cutoff and by which other parameters are measured. Broadly, a battery’s end-of-life (EOL) has been described as the point in time when a given battery can no longer provide enough power or energy to accomplish its intended function [31]. The commonly accepted end-of-life definition of reaching 80 % of original capacity likely stems from the USABC Electric Vehicle Battery Test Procedures Manual, which does not appear to have been updated since 1996 – before modern lithium ion battery powered vehicles were commercially available [23]. As cells have improved, 70–80 % state-of-health for battery EOL has become a common EOL threshold in the literature [28]. State of health (SOH) is typically defined as the percentage of original nameplate capacity (Ah) remaining [32].

However, SOH may also include internal resistance as an additional indicator [32]. In practice, the 80 % threshold is somewhat arbitrary and can occur before catastrophic capacity loss, also called knee-point failure, so lower thresholds of 70 % or 60 % have been suggested [10,30,33]. Battery warranties, such as those from Tesla, replace batteries if the SOH drops below 70 % before the vehicle travels either 100,000 miles or 120,000 miles [29] (SI note 1–6). Note that nominal cell capacity (Ah) should not be conflated with cell energy (Wh) or power (W), as this can lead to misinterpretation of capacity fade. Energy, not capacity, maps to electric vehicle range.

It is important to briefly mention the modal and mechanistic descriptions fundamental to degradation discussion in the literature. Note that this paper focuses on costs rather than materials characterization which is left as a future paper. Three generalized modes are used to describe degradation: loss of lithium inventory (e.g. irreversible lithium plating at the anode, electrolyte side reactions, solid electrolyte interphase (SEI) growth), loss of active material (e.g. binder breakdown, electrode delamination, inaccessible electrode redox) and increased resistance (e.g. SEI growth). For fast charging, irreversible lithium plating is a dominant mechanism. For a given battery system there exists an upper-bound charging rate at which lithium diffusion is inhibited by a mismatch in the available lithium ions and the available electrode sites, given the defined charging time. A gap in the literature discussing mixed fast and slow charger use remains.

We contribute to the literature by providing the first experimental investigation into the relative effect of frequent fast charging on capacity fade, where long duration cycling mixes charging events that are fast or slow. We provide the first experimentally grounded estimates of replacement costs under different amounts of fast charging using BEV-relevant commercial battery chemistries (NMC, NCA, LFP) under behavior-motivated protocols. Protocols are designed to mimic BEV charging patterns ranging from no fast charging (level 2 charging only) to only or extremely frequent fast charging. Thus, to provide experimental insight on the impact of frequent fast charging, this paper investigates the following question: *What is the impact of the number of fast vs slow charging events on estimated BEV battery replacement costs across multiple chemistries?*

Table 2
Summary of notable definitions of fast charging and end of life.

	Source	Definition
Fast Charging	Zhu et al., 2019 [22]	Full charge in 30 min to 2 h ($2C-C/2$) (Extreme fast charging is considered a full charge in less than 10 min or 6C)
	US Advanced Battery Consortium – Electric Vehicle Battery Test Procedures Manual Revision 2 1996 [23]	40 % of battery capacity charged within 15 min, starting from ~40 % SOC “Return 40 % of the state-of-charge of the battery, starting from approximately 60 % depth-of-discharge, in 15 min (i.e. increase from 60 % to 20 % DOD or 40 %–80 % state-of-charge)” (p28)
	National Electric Vehicle Infrastructure program [24]	Power rating ≥ 150 kW
	California Air Resources Board (CARB) [25]	A DC charger which can provide ≥ 23 kW
	Charge Point [26]	“DC fast stations can charge an EV to 80 % in as little as 20 min”
	DOE Alternative Fuel Center [27]	“Approximately 100–200+ miles of range per 20 min of charging ... up to 500 kW ... power varies by vehicle and battery state of charge.”
End of Life	Arrinda et al., 2021 [28]	70–80 % state-of-health (where SOH is capacity/nominal capacity)
	US Advanced Battery Consortium – Electric Vehicle Battery Test Procedures Manual Revision 2 1996 [23]	80 % of original rated capacity or power “When net delivered capacity or peak power capability at 80 % [depth-of-discharge] is less than 80 % of rated” capacity or peak power (p ii)
	Tesla Vehicle Warranty [29]	70 % retention of battery capacity over the warranty time
	Canals Casals et al., 2019 [30]	70 % or 60 % state-of-health (where SOH is capacity/nominal capacity)

2. Experimental procedures

2.1. Battery selection

Panasonic NCA 21700 cells were initially purchased from an online vendor to match those found in a 2019 Model 3 LR Tesla ("2019 M3 LR") as described in SI Note 2–1. The experiment was expanded to include LFP and NMC chemistries; cylindrical 26650 LFP (3.3 Ah, LFP-26650-3300, AA Portable Power Corp) and NMC (5Ah, INR-26650-5000, AA Portable Power Corp), which were purchased from a separate online vendor. Cylindrical 21700 LFP and NCA cells were not readily available so 26650 were substituted to minimize form factor differences that could impact capacity fade. This allows for a more direct comparisons of passive thermal effects and assumption of comparable tab architectures. While Tesla and Rivian BEV models have utilized cylindrical cells, vehicles with LFP, such as the 2022+ Tesla Model 3 Standard Range (272 mi range) use larger format prismatic cells [34]. Experimentation was also limited to a maximum current of 10A provided by the Newareycler.

Purchased cells were randomly selected and cycled (CCCV, 10 mA cutoff) twice at a rate of either C/20 or C/5 to compare to manufacturer specifications and to ensure cell quality. Scanning Electron Microscopy with Energy Dispersive X-ray Spectroscopy (SEM-EDS) suggested the cell chemistries were as follows: NMC 622/Gr, NCA 85 5 10/Gr-SiO₂, and LFP/Gr. Further detail and manufacturer cell specifications can be found in SI Note 2-2.

2.2. Protocol design

Two slow charging cycles were conducted at the beginning of each test to ensure the baseline capacity reference (2nd cycle) used for SOH was not influenced by previous cycling. One batch of cells from each chemistry were cycled in the full manufacturer-provided voltage range (NCA 2.5–4.2 V, LFP 2–3.65 V and NMC 2.75–4.2 V) to approximate a 0–100 % SOC swing. A second batch of NCA and NMC cells were cycled under a restricted voltage range (2.8–3.9 V and 3–3.9 V, respectively) to approximate a 20–80 % SOC swing which is discussed in detail in SI note 2–4. In the battery materials literature voltage can be scaled to model SOC swings [35]. Voltage, or electrochemical potential, can provide more mechanistic information than SOH, or lithiation defined as a percentage with respect to a defined reference.

Slow charging was informed by BEV specifications from the 2019 M3 LR Current for the cells was calculated using a simple model of pack scaling. The nominal pack size for the 2019 T is 78.27 kWh with an onboard AC to DC inverter that has a maximum power output of 11.5 kW [36]. The maximum power for slow charging, also called level 2 charging, is thus 11.5 kW. Dividing these values gives a slow charging rate of 6.8 h which can be extrapolated to C/6.8 and 735 mA for NCA and NMC (5 Ah cells). For LFP, the slow charging current was 634 mA, which is discussed in detail in SI note 2–3. Restricting the useable kWh is one way to derate cells and increase longevity. Thus, the pack energy (kWh) from the cumulative number of cells can be greater than the reported nominal and useable capacity. Defining slow charging using pack to charger power ratio, we model a cell as proportional to a pack. Thus, in the cost model we scale to 4416 cells, leading to an NCA pack of 81.7 kWh vs a 2019 M3 LR nominal energy of 78.27 kWh and a useable energy of 75 kWh. This yields a small overestimation in slow charging current in our protocol design. The final LFP modeled pack was 65 kWh, the NCA pack was 81.7 kWh and the NMC pack was 79 kWh, due to scaling by the number of cells in a pack.

Fast charging was defined as a full charge in 30 min or 2C, which is the capacity normalized unit that relates to current. Fast charging rate was independent of pack size and was kept constant at 2C across chemistries (10 A for NCA and NMC, 6.6 A for LFP). For the 2019 T vehicle a full charge in 30 min is roughly approximated by a charging power of 156.2 kW, near the 150 kW NEVI definition. Upon initial

cycling trials a CCCV 2C charge was found to be longer than 30 min so time cutoffs were added to limit the voltage taper. This means that fast charging represents the maximum amount of capacity that can be moved in the system at a 2C current in 30 min, within the defined voltage range. The time cutoff shortens the time of the CV portion of the CCCV charge compared to the CC portion; in other words, the cell experiences a higher voltage for a shorter period of time than in a traditional CCCV cycle.

Discharging current was based on highway driving which was approximated as 50 miles an hour for a 300-mile pack for NCA and NMC, or a 5-h, C/5 discharge; for LFP a 250-mile pack with a 6-h, C/6 discharge was used.

Five fast/slow charging ratios were chosen to represent a range of charging scenarios: 0, 1/10, 1/5, 1/2 and 25/27. The target ratio of 1 was implemented as 25/27 to allow for test cycles and consistent SOH comparisons. This was because SOH measurements are measured as capacity observed under the slow charging conditions. Capacity varies slightly for cycling utilizing different currents with the same upper voltage cutoff that is not capacity normalized. For simplicity and ease of communication it is referred to as the 100 % fast charging scenario, as the two cases of aging showed knee point onset close enough to not significantly affect the comparative results. See the results section for information on how the five ratios were chosen.

Each Neware cycling protocol can be found in SI note 2–5. Cycling specifications are summarized in Table 3. Data presented in the results section measures state of health against slow charging cycles only.

Table 3
Cycling specifications.

	NCA 21700	LFP 26655	NMC 26650
Cell chemistry	LiNi _{0.85} Co _{0.05} Al _{0.1} O ₂	LiFePO ₄	LiNi _{0.6} Mn _{0.2} Co _{0.2} O ₂
Cell Capacity	5 Ah	3.3 Ah	5 Ah
Supplier	Panasonic	AA Portable Power Corp	AA Portable Power Corp
Discharge Conditions CC	C/6 833 mA	C/5 660 mA	C/6 833 mA
Slow Charge Conditions CCCV	C/6.8 735 mA 10 mA cutoff 6:48 h:min cutoff	C/5.2 634 mA 10 mA cutoff 5:12 h:min cutoff	C/6.8 735 mA 10 mA cutoff 6:48 h:min cutoff
Fast Charge Conditions CCCV	2C 10,000 mA 10 mA cutoff 30 min cutoff	2C 6600 mA 10 mA cutoff 30 min cutoff	2C 10,000 mA 10 mA cutoff 30 min cutoff
Rest between cycles	15 min	15 min	15 min
Full Voltage Range	2.5–4.2 V	2–3.65 V	2.75–4.2 V
Restricted Voltage Range	2.8–3.9 V	NA	3–3.9 V

2.3. Cycling

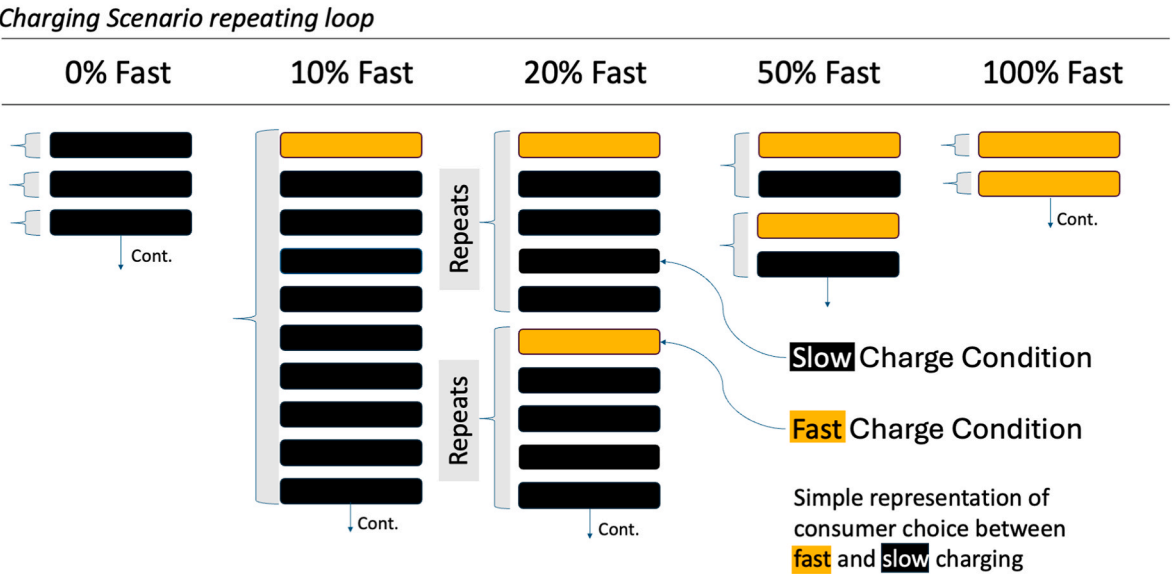
Cells were cycled on a Neware BTS4000-5V10A with CCCV charging and CC discharging under passive thermal management. One cell from each chemistry and SOC condition was monitored with a K thermocouple to record maximum temperature during cycling. Ambient temperature varied from 20 °C to 28 °C. To test for temperature effects, an additional NCA cell was cycled while submerged in ~500 g of paraffin wax, soldered to tabs with a K thermocouple attached via electrical tape to rule out temperature as the dominant degradation mechanism in the NCA 100 % fast charging scenario. See the discussion section and SI Note 2–6 for information on cell temperature during cycling. Cells were cycled long up to 16 months to allow for modeling of at least 150,000 equivalent vehicle miles traveled to represent a typical vehicle lifetime of about 11 years [37].

2.4. Techno-economic modeling using a first order pack approximation

A simple economic model was constructed in R to translate cell degradation findings into implied BEV battery pack replacement costs. Pack replacement occurs when retained battery capacity (SOH) reaches

a defined end of life (EOL) threshold, which we define as 70 % SOH or 150,000 miles, considering a warranty of 100,000 miles (SI note 1–6). Vehicle miles traveled is estimated by multiplying the cumulative energy discharged (Wh) per cell by the number of cells in the pack and a BEV driving efficiency (0.26 kWh/mi) [38]. For the high nickel chemistries the number of cells used was equal to that in a 2019 M3 LR pack, 4416 cells [36]. For LFP we constructed a 65 kWh, 250-mile pack of 6155 cells. As designed the experimental results could be extrapolated to smaller pack sizes where the slow charging rate would map to a lower charging power such that charge time remained constant (SI 2–3). We assume a vehicle is driven 14,000 miles annually and assume the experience of an individual cell is representative of a pack. The model calculates mileage at end-of-life by either matching 70 % SOH to a raw mileage value or uses a weighted average of the two nearest SOH data points. If 70 % SOH was not reached, the model uses a weighted average of the two nearest mileage values to calculate the mileage end of life. Pack costs exclude all non-battery costs, including the BMS, wiring, packaging, installation labor and shipping. We use a discount rate of 5 %. Pack costs are assumed to be fixed at NCA \$172/kWh, NMC \$178/kWh and LFP \$152/kWh based a May 2022 Fast Market analysis [39]. Sensitivities are linear. We do not expect pack replacement costs to

Charging in “Behavior Motivated Cycling”



Common Aging Study Charging

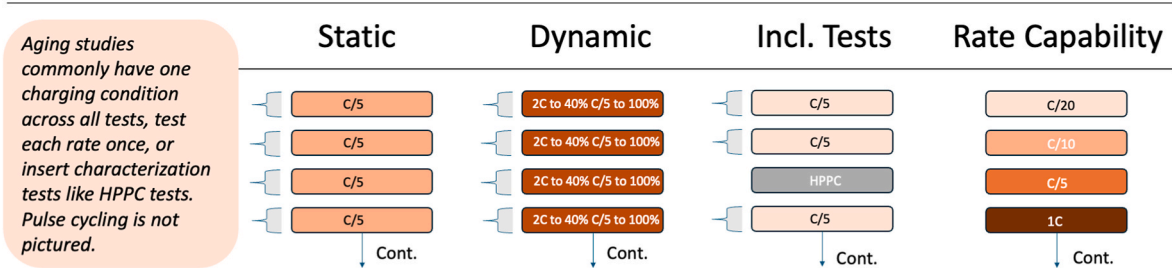


Fig. 1. Illustration of simplistic, behavior motivated charging patterns compared to common aging protocols. Dark blue regions denote cycles with slow charging rates, while yellow regions denote cycles with fast charging rates. Light grey brackets denote what is the repetitive sequence of cycles that characterizes the shortened scenario name. Static refers to a constant rate for the entirety of the charge (e.g. 3C [11]) while dynamic refers to variation in the charge rate during charge (e.g. 2C until 50 % SOC then 6C to 100 % SOC [40]). Dark grey regions denote reference cycling such as high-power pulse cycling (HPPC). Light to dark red regions denote distinct cycling rates. Examples in the literature: Static [11], Dynamic [40], included test [41], rate capability [42]. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

decrease at the same rate as new batteries or packs due to technology lock in.

3. Results

3.1. Protocol design motivation

Because BEV charging can include both fast and slow charger use, we created several charging patterns of different fast-to-slow ratios. The charging rate for a cycle remains the same for the duration of the charge. Our approach differs from common aging studies where the charging rate remains the same across all subsequent cycles (static) or varies within a given charge but remains the same across all subsequent cycles (dynamic), as shown in Fig. 1. Our motivation is to provide structured testing protocols that are more relevant to the kinds of mixed fast and slow charger use that BEVs might experience in practice. For ease of communication, we call this approach “behavior-motivated cycling”.

Using surveys of BEV owners, five different charging scenarios were chosen: 0 %, 10 %, 20 %, 50 % and 100 % of charging events. Specifically, a 2016 survey by Clean Technia, a U.S. media company, found that BEV owners fast charge their EVs at most a few times a month (~10 % or less) [43]. A newer 2024 survey by Flo, a North American BEV charging station company, found that 27 % of respondents primarily used fast charger stations while 31 % relied on a mix of slow and fast charger stations [16]. Since surveys show early adopters fast charged less frequently than recent adopters, accelerated degradation is not

expected to have substantially impacted in the EV market yet. BEV owners with NMC and NCA packs are advised in forums and some EV manuals to fast charge within a 20–80 % range (“the 20–80 % rule”) to minimize degradation. We test experiments under this restriction for the high nickel chemistries (NMC, NCA). Translating scenarios to cycling protocols is discussed later in the section.

3.2. Relative impact of fast charging on battery degradation

Fig. 2 summarizes battery degradation results across full capacity utilization (0–100 % SOC) conditions for all three chemistries. Each testing condition has three tested cells (SI Note 2–7). In battery degradation, there is typically an initial, more rapid decrease in capacity followed by a more linear degradation and finally there is a rapid capacity fade as the cell fails. Capacity loss is generally attributed to loss of lithium inventory, loss of active material, increased resistance, or a combination of all three modes.

The LFP cells show robust capacity retention. After an estimated 150,000 miles traveled the lowest capacity LFP retains roughly 100 % SOH for all scenarios except 100 % fast charging, where it retains 92–97 % SOH after 150,000 estimated vehicle miles. The LFP 100 % fast charging scenario begins to show more significant and accelerating capacity fade after approximately 130,000 miles. LFP’s cycle life is likely supported by its narrower voltage range, contributing to electrolyte stability, and it being a phase change material rather than a layered oxide material, contributing to cathode structural stability. No pack

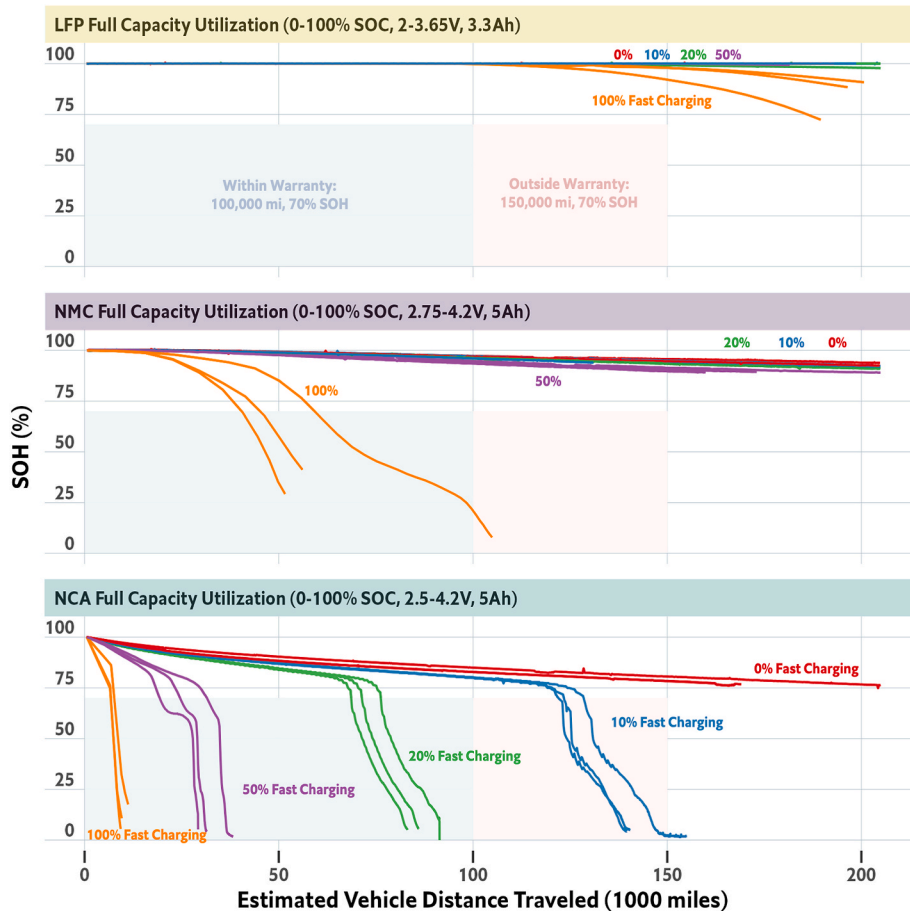


Fig. 2. Effect of fast charging on capacity fade for full capacity utilization. On the x axis, cumulative energy discharged (kWh) of a cell is scaled to estimate the vehicle distance traveled based on a pack made of identical copies of the cell and an assumed efficiency of 0.26 miles/kWh. On the y axis, state of health (% retained capacity) is based on the ratio between the discharge capacity (Ah) of the second reference cycle and any subsequent reference cycle. Data displayed are only from reference cycles to ensure retained capacity, on the y axis, is being compared across the same charging conditions. Cells were cycled to 150,000 equivalent miles for use in the techno-economic model; subsequent cycling data is also included (SI Note 3). See Experimental Section for more information.

replacements are needed to reach beyond 150,000 miles in any other LFP scenarios tested. For the 100 % fast charging scenario, the LFP cells 91 % capacity after 200,000 miles.

The NMC cells last beyond 150,000 miles for scenarios with 50 % or less fast charging but show accelerating capacity fade for the 100 % fast charging scenario. After an estimated 150,000 miles traveled the lowest capacity NMC still has 94–96 %, 94–95 %, and 93–94 % SOH for the 0 %, 10 %, and 20 % fast charging scenarios respectively. After an estimated 200,000 miles, 0 %, 10 %, 20 % and 50 % fast charging scenarios retain 93 %, 92 %, 91 % and 89 % SOH respectively. The For the 100 % fast charging scenario, the NMC cells reach the 70 % SOH threshold at an average of approximately 48,000 miles (40,000–59,000), requiring 2–3 pack replacements to reach 150,000 miles at a cost of \$20,000 to \$32,000. Cost and mileage values have been rounded to two significant figures to emphasize they are modeled values.

The NCA cells show catastrophic failure in less than 150,000 miles in all scenarios that involve any fast charging. After 200,000 miles the 0 % fast charging scenario retains 77 % capacity. Higher fast charging proportions result in earlier failure with observed “knee-point” failure. For the 100 % fast charging scenario, end of life is reached in about 7000 miles (6700–7200), requiring 18–21 replacements to reach 150,000 miles at a cost of \$190,000 to \$220,000. Pack replacement costs, which are triggered at battery end of life increase more rapidly as the portion of fast charging increases.

The observed knee point failure seen in scenarios in Fig. 2 have been suggested to be the result of irreversible lithium plating or from the

accumulation of a critical mass of electrolyte byproducts on the surface of the anode, both of which can be lessened at lower voltages [44]. These byproducts are hypothesized to be formed from electrolyte oxidation reactions at the cathode during cycling, forming more rapidly, at high voltages. The species then migrate from the cathode to the anode where they are reduced to form a layer (the solid electrolyte interphase) on the surface of the anode. After some number of cycles, this layer reaches a critical lack of porosity or thickness, severely hindering kinetics and reducing lithium-ion transport into the bulk of the anode and causing the observed “knee-point” in capacity [45]. Two knee points were observed for the 50 % fast charging scenario, suggesting competition between degradation mechanisms; characterization is left as future work. Note that an SEI also forms at low voltages during cycling but acts as a useful passivating and porous layer in the system.

Fig. 3 summarizes battery degradation results when the NCA and NMC cells are restricted to cycling approximately 20–80 % SOC by restricting the voltage window. SOC as a percentage is set relative to the capacity of a slow charge.

In the restricted case, the NMC cells last beyond 150,000 miles for all amounts of fast charging, but capacity loss is generally higher for more frequent fast charging. After an estimated 150,000 miles, 0 %, 10 %, 20 % and 50 % fast charging scenarios retain 96 %, 94 %, 97 % and 93 % SOH respectively. For the restricted 100 % fast charging scenario, the NMC cells retain 87 % after 150,000 miles.

In the restricted case, the NCA cells last beyond 150,000 miles for up to 20 % fast charging but exhibit a knee point failure and require

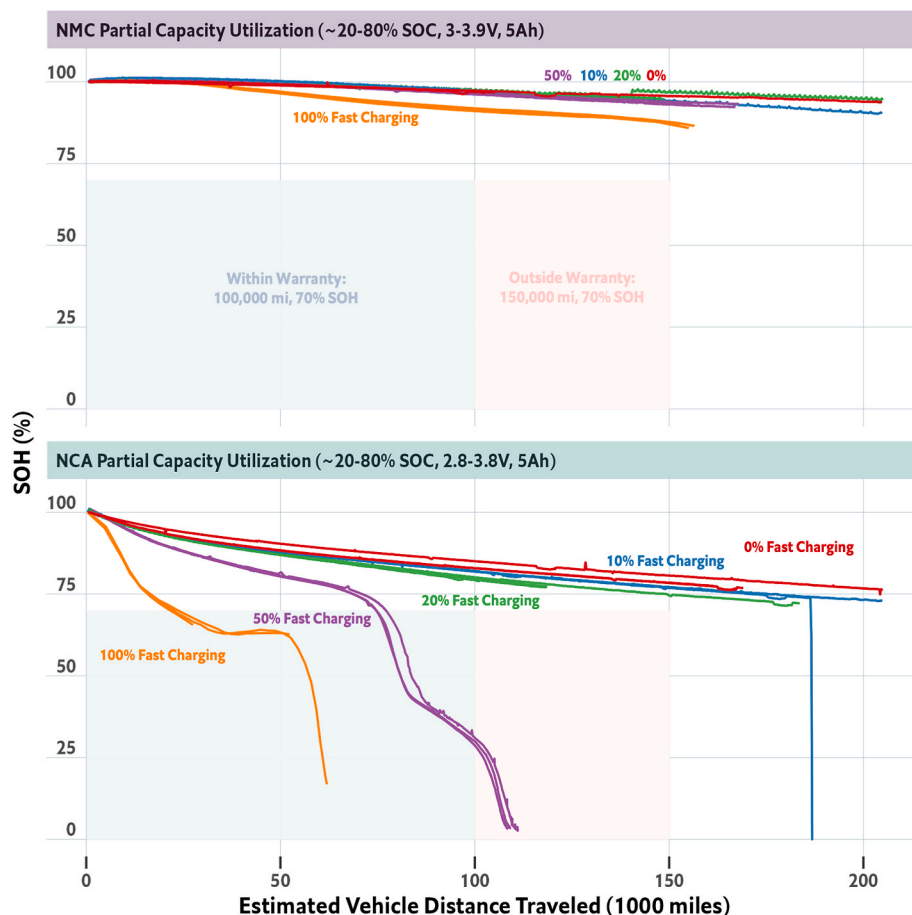


Fig. 3. Effect of fast charging on capacity fade for partial capacity utilization. On the x axis, cumulative energy discharged (kWh) of a cell is scaled to estimate the vehicle distance traveled based on a pack made of identical copies of the cell and an assumed efficiency of 0.26 miles/kWh. On the y axis, % retained capacity (state of health) is based on the ratio between the discharge capacity (Ah) of the second reference cycle and any subsequent reference cycle. Data shown are only from reference cycles to ensure retained capacity, on the y axis, is being compared across the same charging conditions for the cell. See Experimental Section for more information.

replacements prior to 150,000 miles for the two most frequent fast charging scenarios. For the restricted 100 % fast charging scenario, the NCA cells reach the 70 % SOH threshold at around 22,000 miles (21,000–24,000), requiring 6 pack replacements to reach 150,000 miles at a cost of \$62,000 to \$63,000.

Fig. 4 summarizes the state of health of the battery when it reaches end of life at 70 % SOH or after 150,000 estimated miles traveled. Select battery cells were able to cycle beyond 150,000 estimated miles traveled SI Note 3-1 discusses mileage to 80 % SOH and SOH at maximum estimated miles traveled. LFP was not cycled under a 20–80 % SOC condition because (1) our prior results find that LFP batteries do not need to restrict useable capacity to achieve lifetimes well beyond 150,000 miles and (2) the flat voltage profile of LFP makes it more difficult to control with SOC swings with precision.

3.3. Economic impacts of fast charging

Fig. 5 summarizes our estimates of battery pack replacements and their costs over a 150,000-mile vehicle lifetime (~11 years). Pack replacements occur when 70 % of the original capacity remains, based on the warranty of the Tesla Model 3, one of the most popular electric vehicles in the US. Future costs are discounted at 5 %. Pack costs exclude non-battery costs such as installation labor or shipping, and the costs are assumed to be fixed at the following based on a May 2022 Fast Market analysis translating raw material cost to kWh cost estimates: NCA \$172/kWh, NMC \$178/kWh and LFP \$152/kWh. Sensitivities are linear.

To summarize our results, the modeled LFP pack requires no pack replacements beyond a 150,000-mile lifetime, even with the 100 % fast charging scenario and full utilization. In contrast, the same scenario for

the modeled NMC pack requires notable pack replacements (2–3) at cost equivalent to the low end of a new BEV in 2024 (\$20,000–32,000) while modeled NCA pack requires an untenable number of new packs (18–21) at a cost exceeding the typical 2024 BEV price (\$190,000–220,000). The number of required replacements can be reduced for NCA (\$62,000–63,000) and eliminated for NMC (\$0) by restricting capacity usage to ~20–80 % SOC, though this has implications for useable vehicle range and practical energy density, which is discussed in the discussion section.

4. Discussion

An open question in the literature is how to control degradation, which varies across systems and control conditions; these questions should be framed by how much degradation needs to be controlled. Overall, the high performance of the LFP and NMC cells suggests robustness that allows for aggressive operation without severe economic implications. The LFP and NMC cells performed well under passive thermal cooling and high-rate CCCV cycling with very short voltage tapers. That is, frequent fast charging degradation remained low without degradation mitigation measures, such as significant current tapering/dynamic charging, lower voltage cutoffs, and active thermal management. Capacity retention across a majority of the tested scenarios was higher compared to simulations by Neubauer & Wood (2015), which found ~75 %–85 % capacity retention over 10-years, which suggests technology improvements [18].

Our tests use passive thermal management for individual cells in a room temperature environment but active thermal management has the potential to extend battery life [46,47]. As a proxy for improved thermal

State-of-Health at Battery End-of-Life

Cathode Chemistry	Charging Behavior		Estimated Miles Traveled	Avg. Retained Capacity (SOH)
	Capacity Utilization	Fast Charging Scenario		
LFP	Full Utilization 2–3.65V	0%	150,000	100%
		10%	150,000	100%
		20%	150,000	100%
		50%	150,000	100%
		100%	150,000	96%
NMC	Full Utilization 2.75–4.2V	0%	150,000	95%
		10%	150,000	95%
		20%	150,000	94%
		50%	150,000	91%
		100%	48,000	70%
	Partial Utilization 3.0–3.9V ~20–80% SOC	0%	150,000	96%
		10%	150,000	94%
		20%	150,000	97%
		50%	150,000	93%
		100%	150,000	87%
NCA	Full Utilization 2.5–4.2V	0%	150,000	79%
		10%	120,000	70%
		20%	72,000	70%
		50%	25,000	70%
		100%	7,000	70%
	Partial Utilization 2.8–3.9V ~20–80% SOC	0%	150,000	79%
		10%	150,000	77%
		20%	150,000	75%
		50%	75,000	70%
		100%	22,000	70%

Fig. 4. Model results for miles traveled before a battery reaches end of life, either when capacity reaches 70 % of the initial capacity or 150,000 miles are traveled. SOH for select battery cells beyond 150,000 estimated miles traveled and mileage to 80 % SOH are discussed in SI Note 3. Here battery chemistry and charging behavior have a significant impact. Seven out of twenty five tests resulted in a replacement before 150,000 miles.

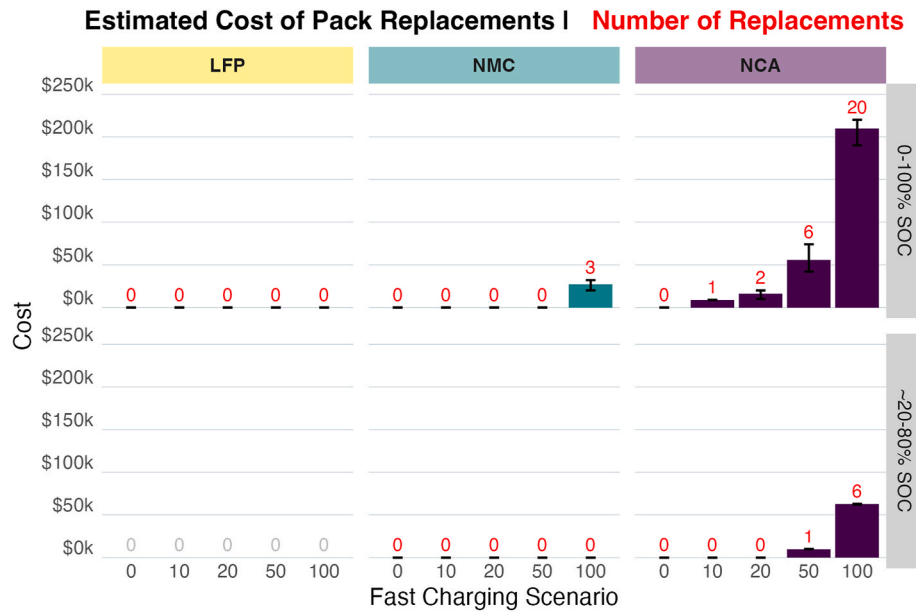


Fig. 5. Present value cost (1000 USD\$) of battery replacements needed to reach 150,000 miles across three chemistries, two capacity utilization scenarios, and five fast charging scenarios for a defined end of life of 70 % retained capacity. LFP 20–80 % SOC is an estimate as it was not observed experimentally but is expected to perform at least as well as 0–100 % SOC. Costs are discounted at 5 %. Values are rounded to two significant digits. Range from min to max span variation across individual cells tested per each condition.

management, we recorded temperature during cycling of an NCA cell embedded in paraffin wax, as NCA had the largest recorded temperature maximum under passive thermal management (72C). While capacity fade was lessened, failure was still rapid and catastrophic such that cycle life remained <75 cycles and battery replacements remained cost prohibitive (see SI Note 2–6 for details).

The tested LFP cells are more resilient to aggressive use conditions and frequent fast charging than NMC and NCA cells, providing an opportunity for comparatively greater practical energy density for LFP packs when minimizing degradation. LFP pack-level energy densities are now competitive with NMC and NCA packs due to LFP's thermal stability and cell design innovations facilitating closer contact between cells [48,49]. Indeed, in vehicle manuals, such as the Tesla's Model 3 Owner's Manual, charging recommendations vary based on battery chemistry. In high nickel (NCA, NMC) battery vehicles, fast charging should be limited to 80 % state of charge (SOC) while for LFP battery vehicles, fast charging can and daily slow charging should charge to 100 % SOC [50,51]. Our results provide evidence to support this 20–80 %

rule for NMC and NCA, but not for LFP. Due to LFP's robustness, the effective energy density of a Tesla with LFP at the real-world pack-level can exceed that of a Tesla with NCA (or NMC) when fast charging, since high-nickel chemistries are restricted to either 20–80 % SOC or 60 % of their rated range to minimize degradation; this pack level difference is visualized in Fig. 6 using range data from fuel-economy.gov.

For the tested NCA cells and NMC cells, the experimental data clearly show that preventing early pack failure for high amounts of fast charging is supported by limiting the percentage of capacity (and energy) cycled, thus limiting the available driving range. These health saving restrictions come at the expense of key factors that BEV purchasers value: range, cost and charging speed [4]. Without high quality BMS control, frequent fast charging could induce large warranty costs or force unreasonable replacement costs on owners who rely on public fast charging, such as those who lack off-street parking for home charging. Lower income households, who can least afford replacement costs, are more likely to inherit older vehicles that reach end of life quickly due to lower quality BMS control and older high nickel chemistries more

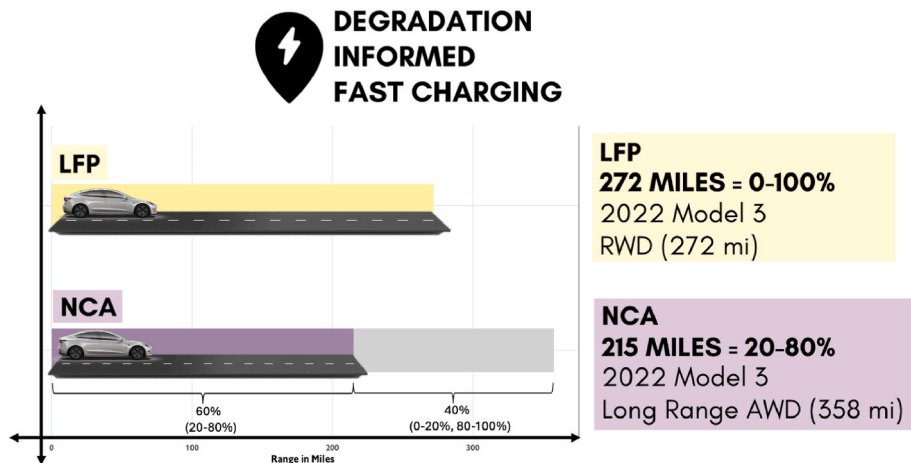


Fig. 6. The EPA range from fuel-economy.gov for the LFP and NCA 2022 Tesla Model 3. NCA is plotted with the 20–80 % rule owners are commonly told to follow to protect batteries when fast charging. Note the LFP Model 3 uses prismatic cells, not cylindrical cells like those investigated in this experiment.

susceptible to fast charge degradation.

In contrast, the experimental data clearly show how LFP cell robustness translates to a larger useable portion of nameplate capacity when fast charging without the risk of early pack replacement. LFP lifetime benefits have also been shown in the literature [10]. Additionally, because LFP can be charged at faster speeds with less effect on life, it provides an opportunity to match battery chemistry with BEV charging use cases: LFP for owners who fast charge frequently.

Despite the advantages LFP shows in our study, LFP cathodes are not without problems. LFP has lower electrical and ionic conductivity as well as more significant path dependency challenges [52]. LFP has a lower theoretical energy density than its high-nickel counterparts. Part of the reason automotive manufacturers recommend regularly charging LFP packs to 100 % SOC is to correct SOC estimation drift. LFP's distinctively flat voltage profile means small voltage variations correspond to large changes in state of charge, rendering voltage-based measurements inherently imprecise and causing state of charge estimations to drift. Power engineers are working to improve management of LFP cells. LFP cell materials remain safer and cheaper to manufacture than nickel-based cathodes with policy further incentivizing a shift in the BEV industry to LFP [53].

4.1. Takeaways for battery electric vehicle owners and policymakers

BEV owners who will need to— or want to — frequently fast charge would benefit from electric vehicles with LFP instead of NCA and NMC battery chemistries. Variation in degradation in our experiment suggests that vehicle mileage alone may not capture the health of a used electric vehicle; fast charging history may be particularly relevant for estimating remaining life in high nickel chemistries. Additionally, because LFP can be charged at faster speeds with less effect on life, smaller battery sizes could still fulfill diverse driving needs as the network of fast charging infrastructure grows. Transporting only a battery as large as needed instead of oversizing for fast charging can increase vehicle efficiency. Matching battery chemistry to usage patterns could further reduce battery waste.

More broadly it is important to keep in mind challenges affecting estimates of pack replacement costs under frequent fast charging conditions. For example, the varying definitions of fast charging and battery end of life, which can create contexts where physical degradation may be masked. The pack energy reported by or used by automakers may be lower than the nameplate energy of all of the cells in the pack assembly. While such de-rating strategies help effectively maintain consistent user experiences to meet warranty requirements, they complicate the assessment of physical cell degradation. This distinction becomes particularly relevant for resource optimization and end-of-life, where accurate degradation metrics are crucial for recycling and second-life applications.

4.2. Limitations

There are several caveats to interpreting our results. Our simple model of frequent fast charging is easy to communicate and repeat, but it may differ from the wide variety of electric vehicle charging cycles that BEVs experience across households. In this experiment, time to fast charge was held constant instead of current or power because this is the measurement that motivates additional BEV sales [3,4]. This provides LFP an advantage in the experimental setup. Its lower volumetric energy density results in cell designs with thinner graphite electrodes (vs NCA, NMC of the same volume), which increases the available anode surface area per unit capacity, reducing local current densities and lowering plating probability. These are functional advantages that are mirrored in the field, where volumetric energy density can be achieved at the pack rather than cell level [48], i.e. with thinner graphite electrodes, but our setup may amplify these advantages.

Industry and electric vehicle manufacturers have access to large field

data sets, which have detailed information on real world degradation. However, due to the short time frequent fast charging has been accessible (specifically 150 kW+), these data rarely capture fast charging battery failure or battery end of life. Note we do not vary the order in which the fast charging occurs (e.g. first 50 % of cycles are fast, the second 50 % of cycles are slow), which is an expected sensitivity in real world operation.

While our study uses simplified cycling to isolate the comparative effects of fast charging across chemistries, our constant-current discharge and constant-current-constant-voltage charging protocols do not capture the complexity of real-world BEV operation, limiting interpretation. The gap in battery lifetime prediction between laboratory based studies and dynamic and/or real-world cycling remains an open question in the literature.

For discharging, real-world BEV driving can include rests, starts and stops as well as rapid accelerations not captured in our protocols. For dynamic discharge protocols, studies have demonstrated conflicting trends. Asean et al. found that dynamic stress test (DST) cycling accelerated capacity loss compared to constant-current discharge cycling for LFP cells undergoing fast charging (2C) [54]. In contrast, Geslin et al. found dynamic cycling showed reduced capacity loss compared to constant-current discharge under the same average current and voltage window for NCA/Gr-SiO₂ cells undergoing slow charging (C/2) [55]. Specifically, the dynamic discharge profile led to an increase of up to 38 % in equivalent full cycles at end of life (85 % SOH) [55]. Note substantial variation in driving patterns across users and applications complicates the selection of representative dynamic duty cycles to characterize or bound capacity loss.

For charging, real-world BEV duty cycles can include charging with significant power tapers and regenerative braking not captured in our protocols. Reducing current at high SOC can increase lifetime as can shortening the amount of time cells experience high current at high voltage [7,56,57].

These findings suggest that while our comparative results remain valid under controlled conditions, the magnitude of degradation observed may vary under real-world conditions. It is more likely than not that real-world degradation is below what is observed in this study. Future work incorporating both dynamic and realistic protocols is needed to understand the gap between lab-based cycling (e.g. CC, CCCV) and dynamic cycling impacts.

Since this experiment began, the commercial company Recurrent Auto published an industry analysis on data it collected from operating BEVs. Recurrent Auto's analysis of 13,000 Tesla vehicles found no statistically significant correlation between fast-charging frequency and range loss; vehicles fast charging more than 70 % of the time did not experience a higher amount of degradation than those fast charging less than 30% of the time. Over 90 % of the vehicles analyzed were 2018 models or newer. Fast charging at or above 150 kW has become significantly more available in recent years [17], suggesting that models older than 2018 were not fast charging as frequently as 70 % of the time.

This differs from our findings for a number of reasons. First, our experimental protocol employs approximately constant-current, in contrast to the dynamic, manufacturer-specific and optimized charging profiles implemented by battery management systems (BMS) in the vehicles studied from the field. The EVSE at which a vehicle is fast charged can also impose control limits. Second, the field study appears to aggregate data across multiple cell chemistries (NCA, NMC and potentially LFP) that our laboratory results indicate have distinct sensitivities to frequent fast charging. Tesla battery chemistries can vary within model years and the analysis does not clarify if they control by model. Third, BMS-imposed charging limitations, particularly the common 80 % state-of-charge restriction for fast charging, may mask degradation patterns observed in more aggressive laboratory testing protocols, especially when combined with de-rating. In our study the 80 % restriction eliminates pack replacements for fast charging for NCA in all but the 100 % and 50 % tests and for NMC for all cases. Furthermore,

the study's primary metric—Real Range Ratio (observed range/EPA range)—may not fully capture capacity fade due to pack de-rating practices. These practices, analogous to battery manufacturers' use of reduced upper voltage bounds to demonstrate extended cycle life, can mask physical degradation. For example, in 2017 during hurricane Irma, Tesla pushed a software update that allowed all owners to access the full 75 kWh, 250-mile range instead of the previously accessible 200 miles [58]. While such strategies effectively maintain consistent user experiences and meet warranty requirements, they complicate the assessment of actual cell degradation. This distinction becomes particularly relevant for resource optimization and end-of-life, where accurate degradation metrics are crucial for recycling and second-life applications.

Finally, while cylindrical NCA and NMC cells are utilized in BEVs directly, LFP chemistries are typically utilized as prismatic or pouch cells. Introducing form factor as another variable expands the parameter space, where attributes like tab design vary more significantly across the tested chemistries, which can impact battery degradation and lifetime. While less realistic, the model cylindrical system was chosen to limit the parameter space, allowing for easier interpretation and supporting future work to develop mechanistic understandings.

5. Conclusion

We contribute to the literature by providing the first experimental investigation into the relative effect of fast charging on capacity fade and battery replacement costs across different fast to slow charging event ratios. The work was motivated by electric vehicle owners who report use of both fast and slow chargers. Cells with BEV-relevant commercial chemistries (NMC, NCA, LFP) are cycled in long term aging studies under the scenarios of 0 %, 10 %, 20 %, 50 % and 100 % fast charging. NMC and NCA cells are also cycled under a restricted voltage condition for ~20–80 % SOC cycling. Data is used in a techno-economic model that assumes a pack is made of identical cells. We report the first experimentally grounded estimates of battery replacement costs under different amounts of fast charging. Real-world degradation and costs can be mitigated via battery management system charging strategies that use lower power or tapered power schedules below the ~150 kW charging rates of our laboratory tests.

The tested LFP cells, under the most frequent fast charging condition and without de-rating, maintained 98 % initial capacity after 150,000 mile equivalent cycling and required no replacements, whereas the NMC and NCA cells dropped below the 70 % SOH (capacity) threshold after approximately 48,000 miles and 7000 miles, respectively, requiring multiple pack replacements to reach 150,000-miles. Thus, the modeled NMC pack requires notable pack replacements (2–3) at cost equivalent to the low end of a new BEV in 2024 (\$20,000–32,000) while the modeled NCA pack requires an untenable number of new packs (18–21) at a cost exceeding the typical 2024 BEV price (\$190,000–220,000). Derating to use only 20 %–80 % SOC extends NMC life beyond 150,000 miles and extends NCA life to 22,000 miles in the 100 % fast charging scenario. Reducing how frequently fast charging occurred generally extended battery life: all derated cells lasted at least 150,000 miles at 20 % fast charging, and all cells lasted at least 150,000 miles with 0 % fast charging (all level 2 charging).

An open question in the literature is how to control degradation, which varies across systems and control conditions; these questions can be framed by how much degradation needs to be controlled. Overall, the high performance of the tested LFP cells suggests a robustness that allows for aggressive operation without severe economic implications. This suggests that policy favoring fast charging will implicitly favor LFP. High-nickel chemistries will require more degradation control to prevent frequent fast charging from inducing unreasonable replacement costs. Fortunately, battery management systems offer complex control solutions (temperature, current, etc.) that are ever improving and can mitigate frequent fast charging replacements of high-nickel batteries.

CRedit authorship contribution statement

H.R. Morin: Writing – review & editing, Writing – original draft, Visualization, Software, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Jay F. Whitacre:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition, Conceptualization. **Jeremy Michalek:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Data statement

Further information and requests for resources and materials should be directed to and will be fulfilled by the lead contact. This study did not generate new unique materials. CSV data for cell retained capacity over time and all original code have been deposited at Figshare and are publicly available at 10.6084/m9.figshare.28811366.v1.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used ChatGPT and Claude to propose alternative words, phrases, sentence structures and shorter title ideas. ChatGPT was also used to support code generation and debugging. Claude was also used to check language clarity and tone. After using each tool/service, the author(s) reviewed and edited the content as needed accuracy and take(s) full responsibility for the content of the published article.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Glossary

NCA	$\text{LiNi}_x\text{Co}_y\text{Al}_{1-x-y}\text{O}_2$
LFP	LiFePO_4
NMC	$\text{LiNi}_x\text{Mn}_y\text{Co}_{1-x-y}\text{O}_2$
EOL	End of Life
SOH	State of Health
SOC	State of Charge
SEI	Solid Electrolyte Interphase

Appendix A Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jpowsour.2025.237552>.

Data availability

<https://doi.org/10.6084/m9.figshare.28811366.v1>.

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